ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B5 · RESOURCES

Renewable or not?

★☆☆
Grade 5

Learning goals

- Classify resources.
- Use the key vocabulary.

1 Sort the resources. [8]

SORT THESE

oil **fish** **wind** **sunlight** **iron ore** **bamboo** **natural gas** **coal**

Renewable	Non-renewable
sunlight	coal
wind	natural gas
bamboo	oil
fish	iron ore

2 Put these in order from fastest to slowest to regenerate (1 = fastest). [3]

1 hilsa population (1–2 years)

3 a mahogany tree (20–50 years)

4 soil (hundreds of years)

5 natural gas (millions of years)

2 bamboo (3–5 years)

3 Match each "R" to an example. [5]

Draw a line to match each pair, or write the letter.

refuse ●	→ E	● **A** refill a water bottle
reduce ●	→ D	● **B** turn old paper into new paper
reuse ●	→ A	● **C** compost vegetable peel
recycle ●	→ B	● **D** turn off the tap while brushing teeth
rot ●	→ C	● **E** say no to a plastic straw

4 **Fill in the gaps.** [3]

WORD BANK

fossil **sustainable** **renewable** **2002** **1971**

Using resources in a way that does not harm future generations is called sustainable use.

Coal, oil and gas are fossil fuels. Bangladesh banned thin plastic bags in 2002

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ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B5 · RESOURCES

Is it sustainable?

★★☆
Grade 5

Learning goals

- Evaluate whether uses of resources are sustainable.
- Explain the impact of human activities.

1 **For each example, decide if it is sustainable and explain why.** [8]

Example	Sustainable?	Why?
Planting a new tree for every tree cut down	yes	the forest regrows as fast as it is used
Catching every hilsa, even the young jatka	no	fish are taken before they can breed, so numbers fall
Using solar panels for electricity	yes	sunlight is renewable and never runs out
Pumping more groundwater than the rain replaces	no	the water level keeps falling

2 **Why is natural gas called non-renewable, even though it was made by nature?** [2]

It took millions of years to form from ancient living things; we use it far faster than it can form again.

3 **Explain how banning plastic bags helps both the environment and people in cities.** [2]

Fewer bags blocking drains means less flooding; less plastic in rivers and soil harms fewer animals; saves oil used to make plastic.

4 **Why does saving vultures matter to people too?** [2]

Vultures eat dead animals, removing a source of disease and keeping the environment clean.

ANSWER KEY

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INVESTIGATE · B5 · RESOURCES

Our class rubbish audit

★★☆
Grade 5

Learning goals

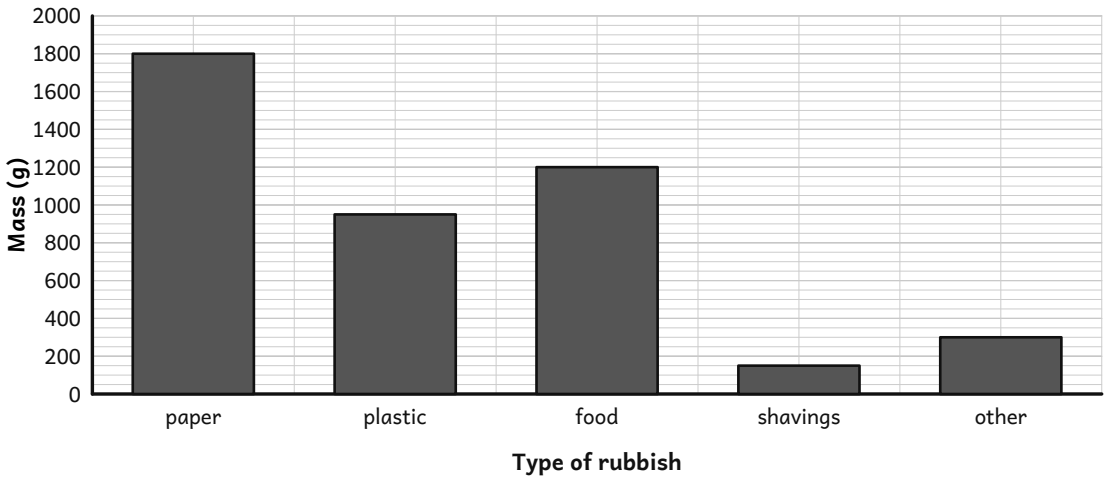
- Collect, sort and record data.
- Present data in a bar chart.
- Use evidence to plan action.

Class 5A collected all the rubbish from their classroom for one week, sorted it and weighed each type.

1 **Complete the table.** [5]

Type of rubbish	Mass (g)	Can it be recycled, reused or composted?
paper and card	1800	yes: recycle
plastic wrappers and bottles	950	some bottles recycle; wrappers mostly no
food waste	1200	yes: compost
pencil shavings	150	yes: compost
other	300	mostly no

2 **Draw a bar chart of the results.** [4]



3 **What was the total mass of rubbish? What fraction was paper and card?** [2]

Total = 1800 + 950 + 1200 + 150 + 300 = 4400 g. Paper: 1800 out of 4400, a bit less than half.

4 **Use the data to suggest two actions the class could take. Say which "R" each one is.** [2]

e.g. use both sides of paper (reduce) and put a paper recycling box in class (recycle); start a compost bin for food (rot); bring tiffin in reusable boxes (reuse/refuse).

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B5 · RESOURCES

Saving the hilsa

★★★ Grade 5

Learning goals

- Explain how rules help a renewable resource regenerate.
- Interpret data over time.
- Consider different points of view.

Science | Economics | Maths | Ethics | Culture

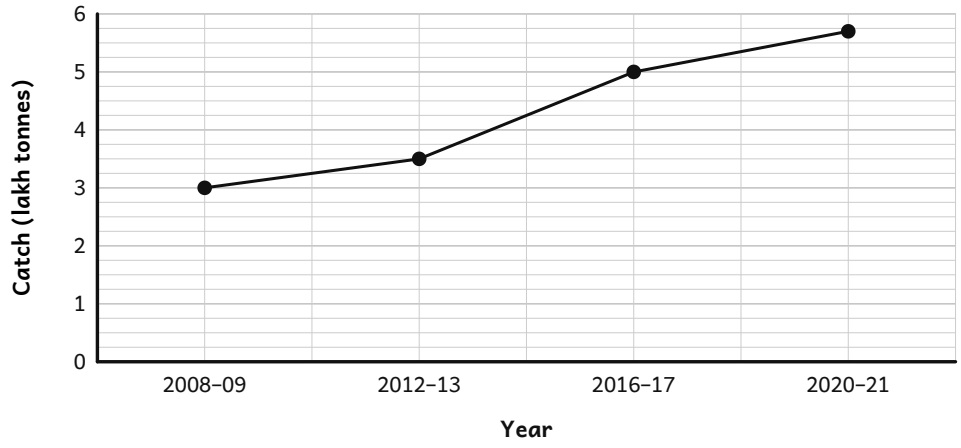
The **hilsa** (*ilish*) is the national fish of Bangladesh. Millions of people depend on catching, selling and eating it. Adult hilsa swim from the sea up rivers such as the Meghna and Padma to lay their eggs. The young fish, called **jatka**, then swim back to the sea to grow.

In the past, so many jatka and breeding adults were caught that the hilsa catch fell. So the government made rules: catching jatka is banned for several months of the year, fishing is banned for **22 days** during the main breeding season in October, and some rivers are **sanctuaries**. During the bans, poor fishing families receive free rice.

Information from the Department of Fisheries, Bangladesh. Catch figures are approximate.

1 **Use the data (approximate hilsa catch in lakh tonnes; 1 lakh = 100 000).** [5]

Year	2008–09	2012–13	2016–17	2020–21
Hilsa catch (lakh tonnes)	3.0	3.5	5.0	5.7



a) Plot the graph and describe the trend.

The catch increased steadily, almost doubling from 3.0 to 5.7 lakh tonnes.

b) Suggest how the fishing bans helped.

More adults could breed and more jatka could grow up, so the population increased.

2 **Why are the bans an example of using a renewable resource sustainably?** [2]

Fish can regenerate if enough are left to breed; the bans give them time to grow back, so people can keep fishing in future.

3 **Complete the table to show different points of view.** [3]

Person	View
a fisher during the 22-day ban	I cannot earn money; the rice helps but is not enough.
a fisheries scientist	The ban lets hilsa breed, so there are more fish for everyone later.
a family buying fish in Dhaka	We cannot buy fresh hilsa for a few weeks, but prices stay lower later.

4 **Suggest one more way to help fishing families during the ban.** [1]

e.g. training for other jobs, small loans, cash support, help to farm fish in ponds or make nets.

ANSWER KEY

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REVIEW AND CHALLENGE · B5 · RESOURCES

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

U	D	R	S	K	R	I	D	R	K	I
K	U	E	V	O	C	S	I	L	B	K
E	Y	D	C	G	B	W	O	S	T	G
Y	D	U	O	R	G	R	F	L	F	N
D	A	C	A	M	E	Y	E	G	A	Y
L	C	E	L	S	O	C	K	U	L	R
W	R	K	N	D	S	U	Y	R	S	T
G	F	O	R	E	S	T	I	C	E	E
I	H	I	L	S	A	G	G	R	L	W
R	E	N	E	W	A	B	L	E	I	E
R	F	C	T	E	U	Y	O	G	A	S

coal **forest** **gas** **hilsa** **recycle** **reduce** **renewable** **reuse** **solar**

2 **Which of these is a renewable source of energy?** [1]

- ○ coal
- ○ natural gas
- ● **wind**
- ○ oil

3 **What does "sustainable" mean?** [1]

- ○ using everything now
- ● **meeting our needs without harming future generations**
- ○ never using any resources
- ○ only using metals

4 **Make a poster plan to persuade your school to save one resource (water, paper or electricity). Include a fact, a picture idea and an action.** [3]

My poster plan

Any clear poster with a correct fact, an image idea and a practical action (e.g. "Turn off fans when you leave").

5 **Challenge. Trees are renewable, yet forests can disappear. Explain how.** [2]

If trees are cut faster than they can grow back (20–50 years), the forest shrinks; renewable resources can still run out if over-used.